Gut microbiome profiling at multiple sclerosis onset as a potential early prognostic marker of disease course: evidence from an observational cohort study

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eBioMedicine
2026;132: 106470
Published Online xxx
https://doi.org/10.1016/j.ebiom.2026.106470

Summary

Background Alterations in gut microbiome composition have been associated with multiple sclerosis (MS), but their impact on disease severity and early progression remains poorly understood. In this study we investigated whether gut microbiome profiling at diagnosis could identify microbial signatures associated with clinical and radiological features of early MS and provide prognostic information.

Methods We analysed the gut microbiome of 53 treatment-naïve patients with MS (pwMS) and 55 healthy donors (HD) using shotgun metagenomic sequencing, combined with clinical features collected over 1 year from diagnosis. To clarify whether gut microbiome composition at MS onset could have prognostic relevance, pwMS were stratified according to lesion burden, lesion localisation, and magnetic resonance imaging (MRI) activity.

Findings Overall beta diversity in Bacteria, Archaea, and Eukarya differed significantly between pwMS and HD (p-value <0.001, <0.02, <0.03, respectively). Within the MS group, glucocorticoid treatment at disease onset was the clinical factor most strongly associated with gut microbiota diversity. Stratification according to lesion burden, lesion localisation, and MRI activity identified two clinically distinct MS subgroups with different baseline clinical characteristics at onset (p-value <0.03) and different risk of early disease progression. The cluster associated with an unfavourable prognosis showed greater progression within 12 months and was enriched for motor symptoms and spinal cord lesions at diagnosis.

Interpretation Our findings suggest that gut microbiome alterations are detectable at the earliest stages of MS and are associated with clinical and radiological features linked to short-term disease evolution. Gut microbial profiling may therefore represent a promising early prognostic biomarker and may help to identify candidate targets for early intervention and therapeutic development in MS, although further validation in larger longitudinal cohorts is needed.

Funding This study was supported by grants from the Italian Multiple Sclerosis Foundation, the Cassa di Risparmio di Torino Foundation, and the Italian Ministry of University and Research.

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Keywords: Multiple sclerosis; Shotgun metagenomic analysis; Gut microbiome; Prognostic biomarker; Disease progression

Research in context

Evidence before this study

To date, numerous studies have established the presence of gut dysbiosis in Multiple Sclerosis (MS). However, the exploration of its impact on disease severity and early progression, especially in treatment-naïve patients at the onset of the disease, is still in its early stages. To investigate this relationship, we conducted metagenomic analysis of the gut microbiome in 53 treatment-naïve people with MS (pwMS) and 55 healthy donors (HD), integrating microbial data with clinical features collected over the first year following diagnosis.

Added value of this study

This study combines shotgun metagenomic sequencing with clinical assessments in a prospective cohort of treatment-naïve pwMS followed for one year from diagnosis. We identified a microbial signature of differentially abundant species related to lesion burden, lesion localisation, and magnetic resonance imaging (MRI) activity, that stratified patients in clusters with different prognosis. These findings link early gut microbiome features with short-term disease progression and prognosis.

Implications of all the available evidence

Our results support the use of gut microbial profiling as a non-invasive tool to stratify pwMS early in the disease course. This approach may enhance prognostic precision, inform early treatment strategies, and provide new targets for microbiome-based interventions. Incorporating microbiome analysis into clinical practice could contribute to more personalised management of MS.

Introduction

Multiple sclerosis (MS) is an inflammatory, demyelinating, and neurodegenerative chronic disease affecting the central nervous system (CNS) and possibly causing severe disability in young adults.[1,2] MS is a heterogeneous disease characterised by variability in clinical symptoms, disease progression rate, treatment response, and underlying several pathophysiological mechanisms. MS can present several phenotypes, with the most common being the relapsing-remitting MS (RRMS), characterised by acute relapses, during which symptoms worsen, alternating with periods of remission.[3] As its prevalence has increased over the past decades, MS has become a significant medical priority, representing one of the leading causes of disability among young adults.[4] Currently, several disease-modifying treatments (DMTs) are available to prevent new attacks and reduce relapse associated worsening. However they differ in terms of efficacy and effects on slowing neurodegeneration.[5] For acute relapses, intravenous steroids remain the primary treatment, but their effect on microbiota is still unknown.

Since the etiopathogenesis of MS is currently unknown, the disease is complex, with both genetic and environmental factors influencing its onset. Among the latter, gut microbiota appears to be a possible influencing element for MS onset and progression. The gut microbiota plays a key role in the gut-brain axis by shaping the immune system and producing hormones and metabolites that influence brain function.[6] Moreover, it has already been reported that the gut microbiota varies between patients with MS (pwMS) and healthy subjects, reinforcing the hypothesis of the role of gut microbiota in the pathology of the disease.[7] Various elements, such as diet or therapies, may influence the microbiota. Several studies have explored gut microbiota changes in MS, focussing mainly on the disease's course and DMT's impact.[8] DMTs can alter the gut microbiota composition, suggesting that their effectiveness may, in part, be attributed to their beneficial impact on microbial balance.[9] Still, few have explored how the microbiome is composed at the onset of the disease in naïve patients[10] and the effects of acute treatments, such as steroids. Finally, many studies use 16S rRNA sequencing technique, which can detect bacterial genera without discriminating between species and excludes domains other than Bacteria. A dysbiosis in Bacteria and Archaea has been reported in MS, and also some yeasts may be implicated.[11–13]

In this study, we investigate the differences in gut microbiota composition between pwMS who never received a DMT, referred to as naïve pwMS at disease onset, and healthy donors (HD). Moreover, we aimed to identify if some differences exist in the gut microbiome of pwMS with an aggressive disease based on negative prognostic factors and evolution in the first year after diagnosis. For this purpose, shotgun metagenomic sequencing was used for an in-depth analysis of gut microbiome, spanning the domains of Bacteria, Archaea, and Eukarya, exploiting Kraken2 workflow.[14] This allowed us to identify gut microbiota alterations at MS onset and provide interesting insights into the effect of glucocorticoid (GC) treatment on the gut microbiome. Moreover, we identified a microbial profile associated explicitly with clinical features and disease evolution, highlighting the potential pivotal role of

the gut microbiota as a potential prognostic factor for disease progression.

Methods

Ethics

This study obtained ethics approval from the ethics committee of A.O.U. San Luigi Gonzaga, Orbassano (TO), Italy (Ref. number #102-2016). Ethics approval encompassed the entire study from different sites. All subjects included in the study provided informed consent to participate. All research was performed following the Helsinki Declaration principles.

Study design and recruitment

For our case–control study, subjects were recruited at four different centres: A.O.U. San Luigi Gonzaga, Orbassano, Italy; Molinette Hospital, Torino, Italy; ASST Spedali Civili di Brescia, Montichiari, Italy; A.O.U. Policlinico Federico II, Napoli, Italy. PwMS were included when receiving a diagnosis of MS according to 2017 McDonald Criteria.[15] For all the subjects, pwMS (n = 69) and HD (n = 18), we considered the following as inclusion criteria: age between 18 and 55 years, an omnivorous diet, a body mass index (BMI) range between 18 and 25, and the ability to express informed consent to the processing of personal data and to store biological samples. Chronic gastrointestinal pathologies, eating disorders, surgery, pregnancy and breastfeeding, and any medical condition that does not allow the signing of informed consent were considered as exclusion criteria. 37 healthy donors (24 females and 13 males) from the Personalised Responses to Dietary Composition Trial (PREDICT1) study (USA)[16] matched by age, BMI and not receiving antibiotic treatment were also included in our analysis. In MS study cohort, demographic, clinical, and MRI data were collected at baseline and for a 12-month follow-up for each subject. For naïve subjects, faecal samples were collected at disease onset, before any MS-specific therapy, except for glucocorticoids (Methylprednisolone 1 g/day intravenously for 5 days). Recruitment has been active from 2017 to 2023, while the experiments and analysis has been conducted from 2023 to 2026.

Assessment of clinical parameters

Demographics (age, sex, BMI) and clinical parameters at diagnosis were collected. Patients with spinal cord lesions, infratentorial lesions, gadolinium-enhancing lesions, and high lesion load at the first MRI were defined as having negative prognostic factors.[17] Disability at diagnosis and during follow-up was defined with the expanded disability status scale (EDSS),[18] ranging from 0 to 10 with 0.5 increments, where 0 indicates a normal global neurological function and 10 death by MS. EDSS assesses various functional systems, with a strong emphasis on ambulation. Baseline MRI scans with gadolinium were performed within 6 months of T0 according to the Italian diagnostic work-up recommendation for clinical practice.[19] Lesion load was reported as low or high according to the Italian Medicines Agency (AIFA) where the threshold of 10 lesions, adopted in major randomised controlled trials[20] was used to define high vs low lesion load.[17] We also recorded GC exposure at baseline and 6 months before faecal sample collection, defining patients exposed or not to steroids. A radiological activity during follow-up was described as new MRI lesions, both symptomatic and asymptomatic, gadolinium positive and not-enhancing, or a new expanding lesion. Relapses, corresponding to a new or worsening neurological symptom lasting at least 24 h, separated by at least 1 month from a previous relapse, were also recorded during the observation period. Finally, disease worsening was defined as one or more of the following: an increase in EDSS score of at least 0.5, new MRI lesions, or clinical relapse events within 1 year.

Faecal sample collection

PwMS and HD were provided with specific stool sample collection kits. Stool samples were self-collected by the patients and HD as previously described.[21] Briefly, the subjects were instructed to self-collect the samples, and all materials were provided in a convenient, refrigerated, specimen collection kit. Patients were given sterile containers to collect the faeces (faecal containers with screw cap 25 ml, with label and spoon, catalogue: #LPIT109147, VWR, Milan, Italy). Faecal samples were collected at home, transferred into sterile sampling containers using a polypropylene spoon (~10 g), and immediately stored at 4 °C. The specimens were transported to the laboratory within 8 h of collection at refrigerated temperature. Containers were immediately stored at −80 °C for DNA extraction. No storage medium was used. Stool samples from external collection sites were shipped frozen with dry ice. All the samples were processed at the Neuroimmunology lab at A.O.U. San Luigi Gonzaga, Orbassano (TO).

DNA extraction from stool and shotgun metagenomics sequencing

Frozen stool samples were thawed and processed using the QIAamp PowerFecal Pro DNA Kit (Qiagen, catalogue: #51804). DNA sample quantity and purity were measured by NanoDrop 1000 (Thermo Scientific™). DNA samples were sent to Novogene Co., UK, for Shotgun Metagenomics Sequencing (Illumina sequencing PE150, 9 Gb raw data per sample).

Metagenomic analysis

Reads from shotgun metagenomics sequencing were aligned to the human reference genomes HG38 and CHM13 using the Burrows-Wheeler Aligner (BWA,

version 0.7.19).[22] BWA is an alignment tool optimised for mapping low-divergent sequences against large reference genomes. Using the HG38 genome (a comprehensive human genome reference) and the CHM13 genome (a complete, gapless reference focusing on difficult-to-sequence regions such as centromeres and telomeres) enabled effective removal of host DNA from the samples. This dual-reference approach ensures greater accuracy by targeting challenging genomic regions, minimising contamination from host sequences in the downstream analysis. Taxonomic assignment, diversity, and differential abundance computation were analysed using the Kraken2 algorithm.[14] Kraken2 is a state-of-the-art tool for taxonomic classification of metagenomic sequences. It uses a k-mer-based approach to assign taxonomic labels by matching subsequences (k-mers) from the query reads to those in a pre-built reference database. We configured Kraken2 (PlusPF database, October 15th 2025) with a confidence score of 0.6, with the aim of reducing false-positives and increasing precision, as described in Wright et al.[23] The taxonomic classification results generated by Kraken2 were further processed using Bracken (Bayesian Re-estimation of Abundance with KrakEN, version 3.1) following the pipeline presented in Lu et al.[24] Bracken refines Kraken2 outputs to provide more accurate estimates of taxonomic abundance. It achieves this by reassigning reads to their most likely taxa based on the k-mer distribution in the reference database, correcting for biases introduced by the uneven representation of taxa in the database. Bracken uses a probabilistic model to estimate the true abundance of organisms at various taxonomic levels (e.g., species, genus, or family). This step ensures that the reported abundances reflect the microbial community composition more accurately, making it a critical component for downstream microbial diversity and differential abundance analyses. Lastly, we used Kraken-biom to transform the Bracken output into the Biom format for the post-process step. The generated Biom was then imported in R (version 4.5.2) and converted into a Phyloseq object for downstream analysis.[25] Phyloseq (version 1.56.0) integrates various microbiome data types, including taxonomic information, abundance matrices, and sample metadata, into a single object, facilitating analysis and visualisation. We first applied DESeq2 (version, 1.50.2)[26] to normalise the data and correct technical artifacts, then we applied a relative abundance threshold of 0.001 to eliminate taxa with minimal representation across samples. This ensures that only the most abundant and biologically relevant taxa remain, filtering out those with negligible presence.

Alpha-diversity and beta-diversity

The Phyloseq object obtained after completing the supervised decontamination step was used to compute the diversities. To assess alpha-diversity, we used the Shannon and Simpson indices (*Vegan* R package) and the richness (Observed) calculated as the number of species present in a sample. Permutational Multivariate Analysis of Variance (PERMANOVA) was performed using the "Phyloseq" R package function. The distance matrix used for the MDS and the PERMANOVA was based on the Bray–Curtis dissimilarity. Comparisons with $P \leq 0.05$ were considered significant.

Differentially abundant species (DAS) analysis

To identify differentially abundant or prevalent species, we employed MaAsLin3.[27] For all comparisons, analysis was conducted against a single variable of interest adjusted for age, sex, BMI and glucocorticoid treatment, when appropriate. Species with a p-value (p) ≤ 0.05 for the first predictor were considered DAS.

Topic modelling for the identification of MS-associated microbial communities

To select the number of latent topics to model, we employed the FindTopicsNumber() function in the ldatuning package, using CaoJuan2009[28] and Arun2010[29] metrics and VEM method for speed. The minimum shared metrics were used to select topic numbers. We employed the LDA[30] function in the topicmodels package[31] to perform the LDA and the tidytext package[32] to extract the per-topic-per-word probabilities via the matrix = "beta" and the per-document-per-topic probabilities via the matrix = "gamma" arguments, both of which we multiplied by read count. Using the linda() function from the LinDA package,[33] we performed differential abundance analysis to test for differentially abundant topics across clinical status and variables, adjusting for sex, age and BMI. We further explored Topics with low FDR p-values (<0.05) and investigated differential abundance among clinical variables of genera belonging to each significant Topic using LDA().

Microbiome functional profiling

Functional read profiling was performed using HUMAnN3 (v3.9)[34] including MetaPhlAn (v201901_v31), DIAMOND (2.0.15),[35] and the databases uniref90 (uniref90_201901b_full)[36] and we merged with Kraken2. To integrate Kraken2 outputs with HUMAnN3, the Kraken2 reports were first converted into a HUMAnN3-compatible format using the kreport2mpa.py utility from the KrakenTools package. This step restructures the taxonomic assignments into a MetaPhlAn-like profile that HUMAnN3 can recognise as a taxonomic input. Since HUMAnN3 requires normalised relative abundances within each domain, we implemented a custom Python script to further adjust the kreport2mpa output. The script reads the Kraken2-derived profile, computes the total number of classified reads at the domain level, and then selects entries

annotated at the genus and species levels. It extracts genus identifiers, calculates their relative abundance as a percentage of the total domain sum, and writes a new taxonomic profile file that includes a header line and genus-level abundances in HUMAnN3-compatible format. This processed file was then provided to HUMAnN3 using the –taxonomic-profile option. Because HUMAnN3 does not accept paired-end files as input, the forward and reverse FASTQ files were concatenated into a single file before running HUMAnN3. The pipeline was finally executed using this combined FASTQ file together with the processed Kraken2 taxonomic profile, ensuring consistency between the taxonomic and functional annotations. After HUMAnN3 completed the taxonomic and functional profiling for all samples, the resulting tables were merged across samples using the humann_join_tables utility provided by HUMAnN3. This step generates consolidated tables of gene families aligning all samples into a single matrix suitable for downstream statistical analyses and visualisation. The joined tables ensure that each feature is consistently represented across samples, facilitating comparative analyses between experimental groups. Reads not mapped to either feature in the databases were counted under the label "UNMAPPED." Similarly, mapped reads that could not be integrated into any pathways were assigned as "UNINTEGRATED." Each pathway is also stratified by taxonomy, labelled "unclassified" if no taxonomy can be inferred. For comparison of samples, the RPK values were normalised to copies per million (CPM) with the human_renorm_table script; then, unmapped, unintegrated, unclassified and species-specific pathways were filtered out. Differential Abundance Analysis (DAA) was performed using MaAsLin 3 (q-value ≤ 0.05).

Setup and execution of the experiment

The workflow implementation is based on the StreamFlow framework,[37] a Workflow Management System (WMS) written in Python and compliant with the Common Workflow Language (CWL) standard. StreamFlow enables concurrent execution of multiple interdependent tasks within a multi-agent ecosystem, facilitating a microservice-based approach. Each task is encapsulated within a Docker image, ensuring modularity and portability. Workflow execution is managed on an High-Performance Computing (HPC) system using SLURM,[38] a highly scalable job-scheduling system that supports the simultaneous execution of multiple workflows.

MS-Explorer: the web application to explore the results

A R Shiny web application, version 1.14.0,[39–41] *MSexplorer*,[42] was developed to facilitate the exploration of results presented in this paper, incorporating various statistical and visualisation techniques, enabling an interactive and user-friendly interface. The dashboard consists of multiple tabs, each designed for a specific analysis: Data Explorer, Principal Component Analysis (PCA), Clustering Analysis, 3D PCA Visualisation, Statistics, Survival Analysis, and Chord Diagram. As input, the application receives a file in which each patient is associated with a corresponding entropy value. The entropy calculation is based on the probability distribution of all possible outcomes of a random variable. In our context, we adapted the formula by computing entropy from the frequencies derived as the ratio between the relative abundance of each species and the sum of the abundances of all species identified as DAS for a given clinical feature. The formula we used is reported below.

$$CF = \{\mathrm{lesionBurden}, \mathrm{spinalcord}, \mathrm{subtentorial}, \mathrm{gadolinium}\}$$

$$DAS_\{cf\} = \{\mathrm{species} \mid \mathrm{species\ showing\ differential\ abundance\ given}\ cf\}$$

$$MS = \{pts_1, \ldots, pts_\{48\}\}$$

$$\mathrm{relAbd}_{\{j\}}^{\{pts,cf\}} = \text{relative abundace of the } j^{\{th\}} \in DAS_{\{cf\}} \text{species in the patient } pts \in MS \text{ computed in } cf \in CF$$

$$H_\{pts, cf\} = -\sum_\{j \in DAS_\{cf\}\} \mathrm{relAbd}_\{j\}\hat{}\{pts, cf\} \log_2\left(\mathrm{relAbd}_\{j\}\hat{}\{pts, cf\}\right), \quad pts \in MS, \; cf \in CF$$

where CF denotes the set of the four clinical features considered in this study, DAS_cf refers to the set of DAS identified in a specific experimental setting, and MS is the list of 48 pwMS. Since only the patients reporting the GC treatment have been selected. For each patient pts, H_pts,cf represents the entropy calculated with respect to a specific clinical feature (cf), where realAbd^{pts,cf}_j corresponds to the relative abundance of the bacteria identified as DAS in that clinical feature for patient pts. Dimensionality reduction (R package *factoextra*, version 1.0.7) was achieved by retaining principal components that explained over 80% of the total variance. Clustering analysis utilised the k-means algorithm (R package *stats*, version 4.4.1) on the reduced data, with an approach for the number of clusters identification.[43] The application evaluated 42 clustering indices to recommend an optimal number of

clusters while providing users the flexibility to adjust cluster count dynamically through an interactive interface. The clustering results are numerically and graphically visualised, including a three-dimensional PCA representation for comprehensive data exploration. Statistical analysis capabilities allow detailed examination of metadata variables within the clusters generated. Researchers could generate summary statistics, visualise mosaic plots, and create contingency tables to investigate relationships within the dataset. Survival analysis is performed using Kaplan–Meier curves, considering any adverse event related to MS progression, including worsening in the EDSS score compared to the EDSS determined at the diagnosis, new lesions found by MRI, or new relapses in the follow-up. The probability of disease progression was estimated using survminer (version 0.5.0) and survival (version 3.7-0). The statistical significance was determined through log-rank testing. Additionally, the application featured a chord diagram visualisation (in-house R function) to illustrate relationships between categorical variables. The chord diagram provides a visual summary of disease evolution regarding clinical worsening events. Flanking the central diagram, two histograms display the distribution of pwMS according to their EDSS scores at diagnosis and 12-month follow-up. The central chord plot illustrates the transition of individual patients between EDSS score categories. Ribbons between segments represent the flow of patients from one EDSS category to another, where the width of each ribbon is proportional to the number of individuals undergoing that specific transition, becoming proportional to the number of individuals in that EDSS category and cluster. This visualisation can capture the initial distribution and the evolution of patient status over time. Colours are assigned based on the patient clusters identified through the stratification analysis, allowing for immediate visual association between prognosis groups.

Statistics

All statistical analyses and visualisations presented in this study were performed using the R programming language, version 4.4.3. For categorical variables, the p-values were computed using the Chi-Square or Fisher's exact test, with the latter applied for 2x2 contingency tables or when expected frequencies were below five. For comparisons involving a quantitative variable across two groups defined by a qualitative factor, statistical tests included the Kruskal–Wallis test for median differences, the T-test or Wilcoxon-Mann-Whitney test for mean comparisons (based on normality and sample size), and the F-test or Levene's test for variance equality. The Kolmogorov–Smirnov test was used for distribution comparisons when sample size permitted. For comparisons across more than two groups, the Kruskal–Wallis test, ANOVA, Bartlett's test, or Levene's test were used depending on normality and homogeneity of variance assumptions. All p-values are nominal p-values. Linear regression analyses were conducted using the leaps package (version 3.2), which implements an exhaustive search for the optimal subsets of predictor variables (X) for modelling the response variable (Y) in linear regression. The selection process is based on a computationally efficient branch-and-bound algorithm.

Role of funders

Funders played no role in the design of the study, the collection of data, the analysis and interpretation of data, or the writing of the manuscript.

Results

Diversity and abundance estimation of bacteria, archaea and eukarya in pwMS and HD

We analysed the gut microbiome of 124 subjects divided into pwMS (n = 69) and HD (n = 55). Most pwMS were enrolled at the time of disease onset and were treatment-naïve at the time of sample collection (n = 58, 84.1% of the total) excluding the non-naïve. The characteristics of the subjects enrolled are reported in Supplementary Table S1. Five out of 58 (7.2%) pwMS used antibiotics in the three months before sample collection and were excluded from further analysis. The gut microbiome of the remaining 53 pwMS was further analysed as a naïve MS study cohort, in addition to the cohort of 55 healthy subjects. Participants followed an omnivorous diet and exhibited moderate adherence to the Mediterranean diet (45.15 ± 11.79), as assessed by the Pyramid-based Mediterranean Diet Score (PyrMDS) questionnaire.[44] Naïve-MS and HD ages were similar, with a mean age of 36 ± 10 for the HD group and 35 ± 10 for the naïve-MS group. The sex distribution of subjects was as follows: 33 (60%) females and 22 (40%) males in the HD group and 32 (60.4%) females and 21 (39.6%) males in the MS-naïve group. All subjects were within the normal weight range, which is defined as a Body Mass Index (BMI) between 20 and 25. The baseline EDSS score of pwMS ranged from 0 to 4. The most frequent EDSS score was 1 (n = 13), followed by EDSS 0 (n = 11) and EDSS 2 (n = 10). Lower frequencies were observed for EDSS 1.5 (n = 6), 2.5 (n = 2), 3 (n = 2), 3.5 (n = 2), and 4 (n = 1). Radiological examination with MRI classified naïve-MS based on MRI clinical features at diagnosis.

The distribution of lesion burden was as follows: 28 (54%, n = 52) subjects had a low lesion burden (<10 lesions), while 24 (46%, n = 52) had a high lesion burden (≥10 lesions). Lesion localisation and activity were also evaluated, revealing that 27 (55%, n = 49) MS-naïve had spinal cord lesions at diagnosis, while 22 (45%, n = 49) did not; 20 (41%, n = 49) subjects had subtentorial lesions, whereas 29 (59%, n = 49) did not;

and 14 (29%, n = 49) of MS-naïve had Gadolinium-enhanced lesions at baseline, while the 35 (71%, n = 49) did not. GC (methylprednisolone 1 g/day for 5 days) was administered to a subset of MS-naïve to alleviate acute phase symptoms. Consequently, 22 (46%, n = 48) naïve-pwMS received GC treatment within the six months before sample collection, while 26 (54%, n = 48) did not.

Descriptive statistics of the study cohort are summarised in Table 1. Demographic and clinical characteristics stratified by sex are reported in Supplementary Table S2.

Microbial DNA was extracted from stool samples collected from HD and pwMS at time of disease onset. The extracted DNA was subjected to shotgun metagenomic sequencing, and microbial taxonomic profiles, including taxa from the Bacteria, Archaea, and Eukarya domains, were generated using the workflow outlined in Fig. 1. Details of reads mapping are reported in Supplementary Table S3. In total, we identified 259 bacterial species across 8 phyla, 5 archaeal species across 2 phyla, and 10 eukaryotic species across 3 phyla (see Supplementary Table S4 for complete list of microbes and their abundances). The ten most abundant species within each domain (Bacteria, Archaea, and Eukarya) are presented in Fig. 2a, d, and g, respectively. We found a significant association between microbial richness and MS status when comparing pwMS and HD across all domains. Alpha diversity (Observed, Shannon, and Simpson) did not reveal consistent differences between pwMS and controls across Bacteria (Fig. 2b), Archaea (Fig. 2e), and Eukarya (Fig. 2h), except for higher Observed richness in Bacteria and increased Shannon index in Archaea in pwMS (see Supplementary Table S5 for full list of alpha diversity indices). No effect of diet on alpha diversity was detected (all comparisons non-significant, Supplementary Figure S1). PCoA was performed to assess overall differences in microbial community composition between HD and pwMS. PCoA based on Bray–Curtis distances revealed significant compositional differences in gut Bacteria, Archaea, and Eukarya between study groups (Fig. 2c, f, and i).

DAS in pwMS with respect to HD were identified using MaAsLin3 and adjusting for key covariates—including age, sex, and BMI—to account for potential confounding effects. Specifically, 71 Bacteria, one Archaea and two Eukarya were identified and listed in Supplementary Table S6 (MaAsLin3 results) and represented in Fig. 3a. Specifically, in the phylum Bacillota, notable increases were observed in members of the family *Peptostreptococcaceae*, including *Romboutsia ilealis*, *Mediterraneibacter [Ruminococcus] torques* belonging to the family of *Lachnospiraceae*. Moreover, in the phylum Actinomycetota, significant increase was observed for *Collinsella aerofaciens* and *Bifidobacterium* genus, including *B. adolescentis*, *B. longum* and *B. catenulatum*. Among other domains we observed an increase of *Methanobrevibacter sp. TLL-48-HuF1* for the Archaea and of *Komagataella phaffii* and *Saccaromyces cerevisiae* for the Eukarya domain.

Then, we applied topic modelling to the microbiota analysis to identify recurring bacterial community types, which are not captured when considering individual taxa in isolation but instead reflect patterns of co-occurrence among genera that may be relevant for differences between pwMS and HDs. We selected 12 topics as they had the lowest metrics (Supplementary Figure S2a) and performed differential abundance analysis to test for differentially abundant topics across clinical status, while adjusting for sex, age and BMI. Among the 12 topics selected, we identified 2 topics with significant FDR p-values (<0.05) and we examined the genera with high probabilities of assignment to these topics, i.e. community types (Supplementary Figure S2b). Topic 8 was characterised by a high probability of assignment of *Roseburia* (Supplementary Figure S3a, left panel), which is increased in pwMS with respect to HDs (Supplementary Figure S3a, right panel), and *Faecalibacterium*, which is stable across donor groups. In topic 11, *Bifidobacterium*, which is significantly increased in pwMS compared to HDs (Supplementary Figure S3b, right panel), was the most often assigned genus, followed by 19 genera with low probability of assignment (Supplementary Figure S3b, left panel). Among the above-described genera, several species were found to be differentially abundant in our pipeline as well, confirming our findings.

Finally, we performed functional metagenomic profiling identifying differentially abundant pathways between HD and pwMS, that include amino acid metabolism (histidine, methionine, cysteine, glutamine and phenylalanine), fermentation and respiration, including pyruvate conversion routes leading to acetate, lactate, and isobutanol, as well as succinate fermentation to butanoate (see Supplementary Table S7a). Moreover, also carbohydrate and complex sugar metabolism (chitin, lactose, and galactose degradation and biosynthesis of fructans, UDP-glucose, UDP-N-acetylglucosamine), nucleotide metabolism (biosynthesis and degradation of pyrimidines), and cofactor and lipid pathways (biosynthesis of factor 420 and of geranylgeranyl-PP via the mevalonate pathway) resulted modulated (Fig. 3b and Supplementary Table S7b). Interestingly among reduced pathways we found succinate fermentation to butanoate, suggesting a diminished microbial capacity for butyrate production, a short chain fatty acids (SCFA) with well-established anti-inflammatory and neuroprotective properties.

Microbiome in the gut of pwMS is affected by glucocorticoids treatment

We performed a regression subset selection analysis of clinical variables using a linear regression model, identifying GC-treatment as the most relevant clinical

Characteristic	N	HEALTHY N = 55[a]	MS N = 53[a]	p-value
Age	108	36 ± 10 [22, 54]	35 ± 10 [19, 55]	>0.9[b]
BMI	103	22.27 ± 1.71 [18.00, 24.99]	23.17 ± 3.37 [16.80, 32.10]	0.10[b]
Sex	108			>0.9[c]
F		33 (60%)	32 (60%)	
M		22 (40%)	21 (40%)	
EDSS diagnosis	47			
0			11 (23%)	
1			13 (28%)	
1.5			6 (13%)	
2			10 (21%)	
2.5			2 (4.3%)	
3			2 (4.3%)	
3.5			2 (4.3%)	
4			1 (2.1%)	
EDSS progression	41			
0			8 (20%)	
1			11 (27%)	
1.5			5 (12%)	
2			9 (22%)	
3			3 (7.3%)	
3.5			3 (7.3%)	
4			2 (4.9%)	
Gc treatment	48			
Untreated			26 (54%)	
Treated			22 (46%)	
Lesion burden	52			
High			24 (46%)	
Low			28 (54%)	
Spinal cord lesion	49			
Yes			27 (55%)	
No			22 (45%)	
Subtentorial lesions	49			
Yes			20 (41%)	
No			29 (59%)	
Gadolinium contrast	49			
Yes			14 (29%)	
No			35 (71%)	
Clinical presentation	49			
Spinal cord			6 (12%)	
Brain stem			7 (14%)	
Cerebellar			1 (2.0%)	
Motor impairment			6 (12%)	
Multifocal			2 (4.1%)	
Optic neuritis			11 (22%)	
RIS			2 (4.1%)	
Sensory symptoms			14 (29%)	

The study cohort consisted of healthy donors (HD, n = 55) and treatment-naïve people with multiple sclerosis (pwMS, n = 53). Continuous variables are presented as mean ± standard deviation (SD) [range], and categorical variables are reported as n (%). BMI = Body Mass Index. EDSS = Expanded Disability Status Scale at diagnosis (EDSS diagnosis) or after one year of follow-up (EDSS progression). The numbers below correspond to the disability scores assigned to patients according to the EDSS. RIS = Radiologically Isolated Syndrome.
N = number of subjects with available data for the corresponding feature. [a]Mean ± SD [Min, Max]; n (%).
[b]Wilcoxon rank sum test. [c]Fisher's exact test.

Table 1: **Baseline demographic and clinical characteristics of the study cohort.**

factor influencing gut microbiota diversity in the MS group, among sex, age, BMI, MRI lesion burden, and lesion localisation (F-statistics p-value = 0.0094), suggesting that GC treatment is a potential strategy for patient stratification. Thus, we compared the gut microbiota of the 22 GC-treated pwMS with the 29 GC-untreated pwMS in our cohort, finding that GC-treatment did not affect the alpha-diversity of Bacteria (Fig. 4a) and Eukarya (Supplementary Figure S4b), while reduced Observed richness and Shannon index was reported for Archaea (Supplementary Figure S4a). Beta-diversity of all the domains (Fig. 4b and Supplementary Figure S4c and d) were not statistically different between GC-treated and GC-untreated pwMS.

Considering the influence of GC treatment on gut microbiota diversity, we performed a differential abundance analysis comparing GC-treated and GC-untreated pwMS. We identified 22 Bacteria and no Archaea or Eukarya (represented in Fig. 4c and listed in Supplementary Table S6b). Bacterial species belonging to Bacteroidota, Pseudomonadota and Bacillota phylum were found to be differentially abundant. Among bacteria reduced by GC-treatment we found species belonging to *Streptococcaceae (Streptococcus salivarius and Streptococcus vestibularis)* and *Sutterellaceae (Sutterella wadsworthensis, Duodenibacillus massiliensis) and Veillonellaceae (i.e., V. dispar, V. nakazawae, V. parvula)*, while *Phoecicola dorei* and *Paraprevotella clara* were found increased.

Establishment of microbial species associated with clinical features

We aimed to identify DAS associated with specific clinical features selected for their clinical relevance. To achieve this, we performed differential abundance analysis on bacterial taxa and adjusting for GC treatment to stratify patients based on four distinct clinical features: lesion burden (Lesion high/Lesion low), presence of gadolinium-enhancing lesions (Gad+/Gad-), spinal cord lesion localisation (Spinal positive/Spinal negative), and infratentorial lesion localisation (Infratentorial positive/Infratentorial negative) (see Supplementary Table S8 for full lists).

The four bacterial species lists, comprising a total of 46 DAS, are available on the MSExplorer Shiny R application (https://qbiogroup.shinyapps.io/MSexplorer/). Fig. 4d presents a heatmap of these 46 species, illustrating their associations with clinical features, with colour intensity reflecting the corresponding model coefficients. The majority of species exhibited specificity for a single clinical variable; notably, *Faecalibacillus intestinalis* represented the only exception, displaying differential abundance across both clinical features and lesion localisation.

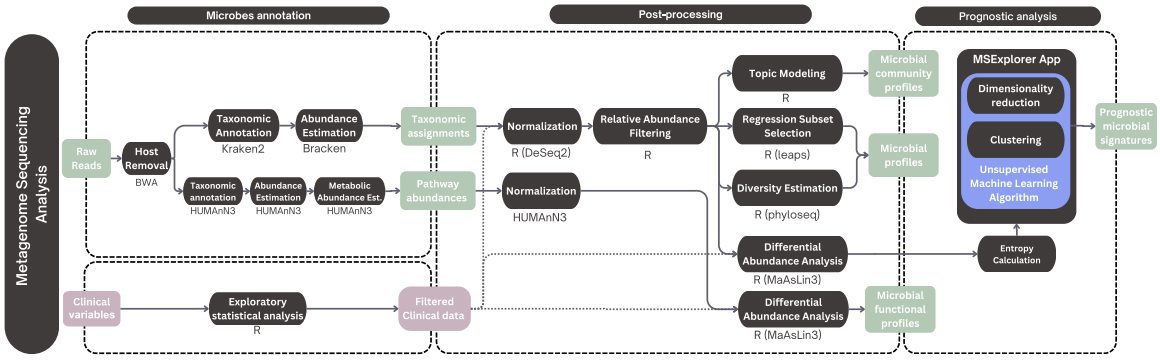


Fig. 1: **Workflow of analysis.** The metagenome sequencing analysis workflow consisted of three parts. The first part is "Microbes Annotation", in which raw reads from shotgun metagenomic sequencing are first aligned to the human reference genome to remove host DNA. The remaining reads are then assigned to microbial taxa using Kraken2 and Bracken and pathway abundance is estimated using HUManN3. In parallel, sample metadata is processed for exploratory statistical analysis. The second part is "post-processing", in which taxa-annotated counts are normalised using DESeq2 and subjected to relative abundance filtering. This is followed by differential abundance analysis to identify differentially abundant species, diversity estimation to compute alpha and beta diversity metrics, topic modelling analysis to discover microbial communities, and regression subset selection to identify associated clinical features. HUMAnN3 output is normalised and analysed for differential abundance. The third part is "Prognostic analysis", in which the entropy indices of the processed microbial profiles are analysed by unsupervised machine learning algorithms (i.e. PCA and k-means clustering). The results are then visualised using the MSexplorer Shiny R app.

Among the identified taxa, six species demonstrated both a statistically significant association with clinical variables and a consistent differential abundance between HDs and pwMS (Fig. 4e). Specifically, *Bacteroides ovatus* and *Bacteroides cellulosilyticus* were depleted in the pwMS cohort and were further reduced in high lesion burden and subtentorial lesion localisation, respectively. Conversely, *Barnesiella sp. An22* and *Romboutsia ilealis* were enriched in pwMS and correlated with increased lesion burden. Similarly, *Clostridium innocuum* and *Ruminococcus sp. JL13D9* were elevated in pwMS and associated with spinal cord lesion localisation.

In light of prior evidence linking these taxa to MS, the observed patterns suggest that the depletion of *Bacteroides* species may indicate a loss of anti-inflammatory and neuromodulatory functions. In contrast, the enrichment of *Clostridium innocuum* and *Ruminococcus* species is more consistent with a state of inflammatory dysbiosis.

Then, we performed topic modelling analysis for MS subjects, as previously described for healthy vs pwMS comparison. We selected 7 topics (Supplementary Figure S5a), but found no statistically significant difference in community abundance across clinical variables, while adjusting for GC treatment, sex, age and BMI (Supplementary Figure 5b–e).

Assessment of the bacterial profile as an early prognostic biomarker of MS progression

In this section, the focus of the paper shifts from feature-specific microbial signatures to community-level structure. Specifically, in this phase, we aimed to interpret microbial variation at the community level rather than relying on individual species. Each set of DAS obtained considering a given clinical trait was used to compute entropy values (see Methods Section for more details). Subsequently, an unsupervised dimensionality reduction approach was applied to the alpha-indices, employing PCA to represent each pwMS in a reduced biodiversity feature space, bringing together subjects with similar abundance profiles (Fig. 5a). Notably, the extended engineering vectors (blue arrows in the scatter plot) indicate that clinical features contribute substantially to the variance captured by the first principal component. The orientation of these vectors further reveals positive correlations among lesion localisation variables, as well as between lesion count and lesion activity. In contrast, an inverse relationship is observed between lesion localisation and both lesion number and activation status.

Following PCA, we performed k-means clustering to identify patient subgroups and statistically evaluated clinical features across the identified clusters. To meaningfully stratify the patient cohort, we applied k-means clustering with $k = 2$, as determined optimally by consensus across 15 clustering indices (see Methods Section). This approach successfully delineated two distinct patient groups.

To evaluate whether the identified bacterial species are associated with the progression of clinical course, we validated the clusters by assessing their ability to discriminate between individuals who experienced worsening clinical outcomes within 12 months of diagnosis. In the survival analysis, events are defined as one of the following: an increase in EDSS score of at

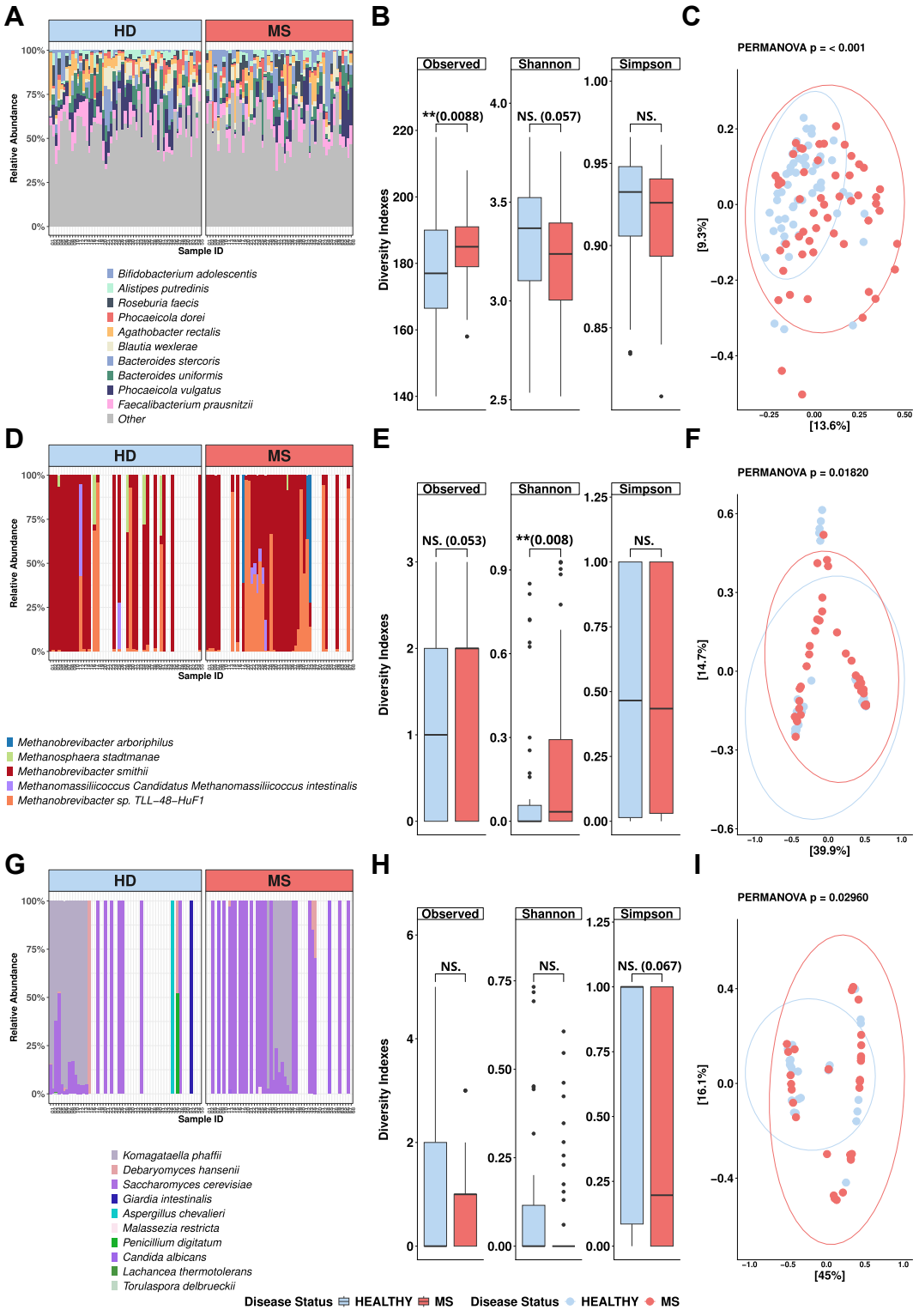


Fig. 2: **Gut microbiota composition and diversity in PwMS and HD. A–I** Stacked bar plots representing the top 10 most abundant bacteria (**A**), archaeal (**D**), and eukaryotic (**G**) species in the study cohort (55 HD and 53 PwMS). Alpha-diversities are indicated as Observed, Shannon and Simpson indices in HD (light blue) and pwMS (red) for Bacteria (**B**), Archaea (**E**) and Eukarya (**H**) Wilcoxon rank-sum test $p > 0.05$ NS, not statistically significant. Data are represented as boxplots. Beta-diversities computed in HD and MS groups for Bacteria (**C**), Archaea (**F**) and Eukarya (**I**). All comparisons were statistically significant; (PERMANOVA, $p < 0.05$) The percentages reported in square brackets next to the PCoA axes indicate the proportion of the total variation in the distance matrix represented by each principal coordinate.

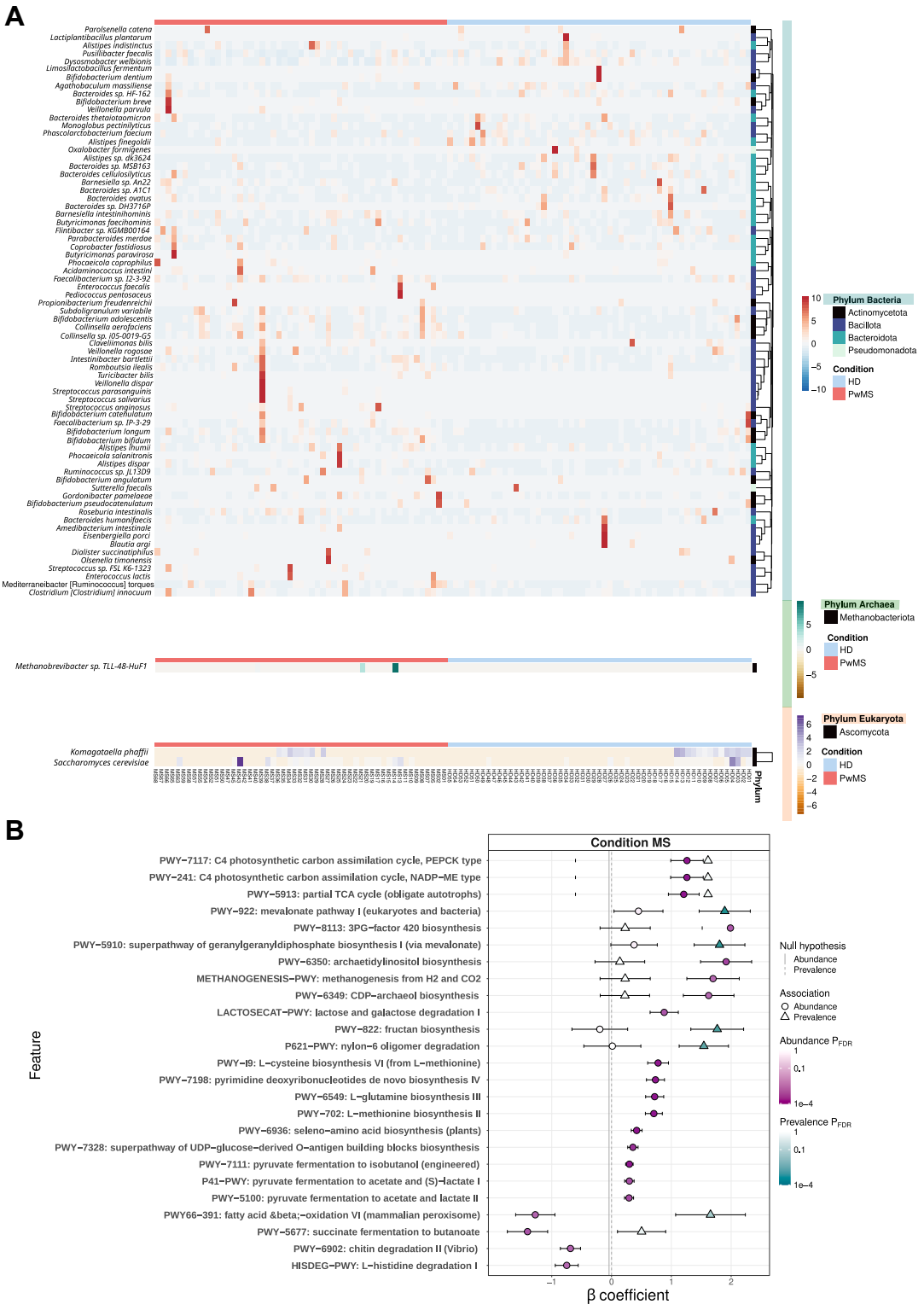


Fig. 3: **Differential abundant species and pathways identified in the comparison between pwMS and HD. A**. Heatmap representing DAS identified among Bacteria, Archaea, and Eukarya between 53 pwMS and 55 HD. The heatmaps highlight the DAS obtained in Bacteria (top), Archaea (middle), Eukarya (bottom) domains. Heatmaps were generated by grouping subjects based on the disease status. Row clustering was performed using species distance metrics within each domain. The column header represents disease status, whereas the row headers indicate the phylum to which each species belongs. **B**. Differentially abundant pathways between MS and HD identified using MaAsLin3. The figure displays the top 20 pathways significantly associated with disease status, with effect sizes and directions indicating increased or decreased abundance in MS (Benjamini–Hochberg FDR correction; q-value <0.05).

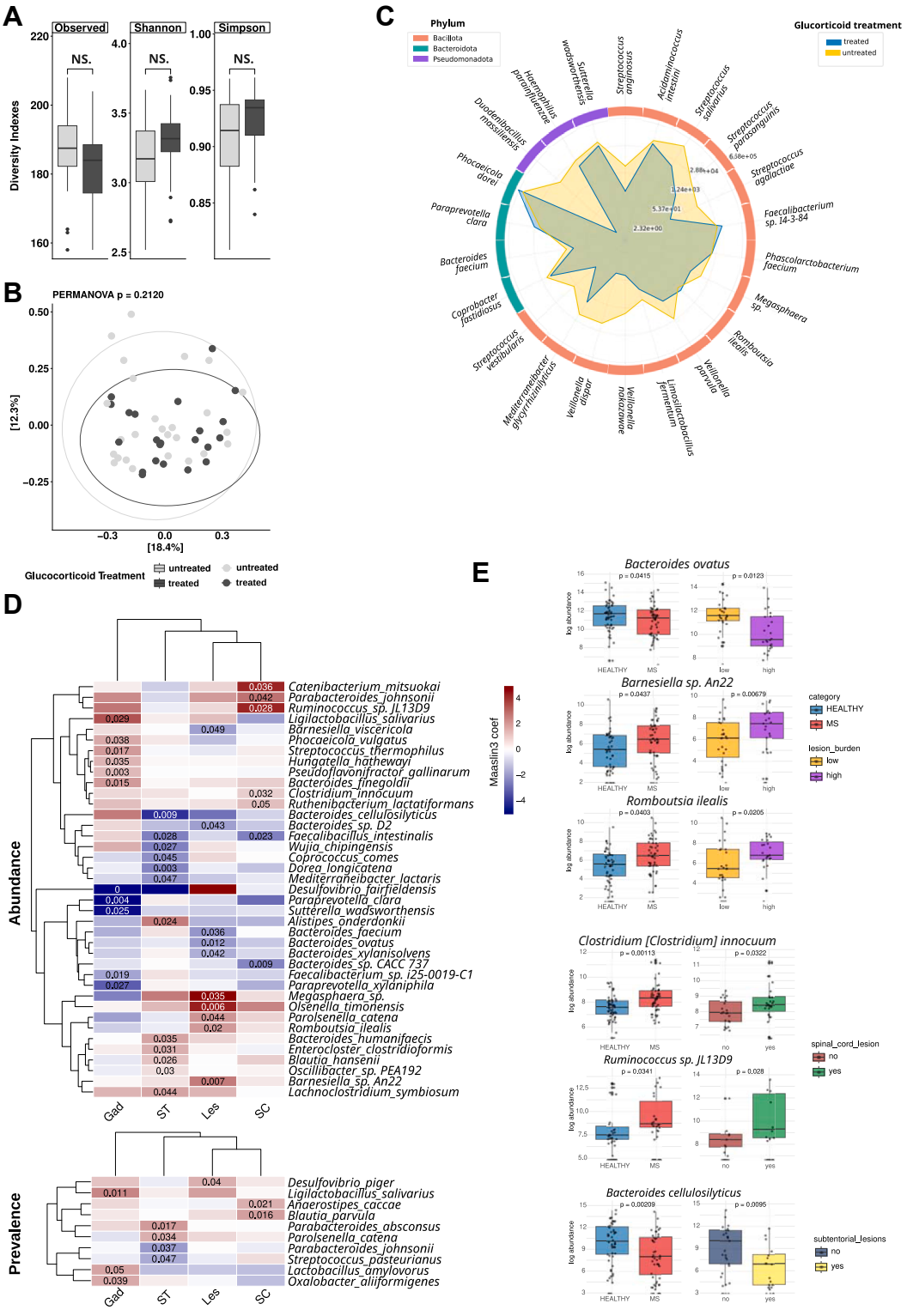


Fig. 4: **Establishment of microbial species associated with clinical features. A**. Alpha-diversities evaluated with observed species, Shannon and Simpson diversity indexes among Bacteria in 22 GC-treated and 26 GC-untreated pwMS. Wilcoxon rank-sum test; p-values >0.05 NS (not significant), ≤ 0.05 *, ≤ 0.01 **, ≤ 0.001 ***. **B**. Beta-diversities assessed with PCoA based on Bray–Curtis distance among Bacteria in GC-treated and GC-untreated pwMS. PERMANOVA test; p-values >0.05 NS (not significant), ≤ 0.05 *, ≤ 0.01 **, ≤ 0.001 ***. The

least 0.5, the appearance of new MRI lesions, or relapse within 1 year. The Kaplan–Meier plot (Fig. 5b) revealed that patients assigned to the yellow cluster, indicative of a favourable prognosis (FP), had a non-significantly lower risk of early progression (log-rank p = 0.065) than those in the burgundy, indicative of an unfavourable prognosis (UFP).

Furthermore, the patient clustering analysis yields a notable and clinically relevant finding. The two identified clusters exhibit marked differences not only in microbial composition and disease progression trajectories, but also in baseline clinical characteristics at diagnosis. Specifically, Fisher's exact test (p-value = 0.03) reveals a significant enrichment of motor impairment symptoms and spinal cord lesion localisation in the cluster associated with earlier disease progression (Fig. 5c).

These results are consistent with, and further support, existing evidence indicating that both motor impairment at onset and spinal cord involvement are associated with accelerated disability accumulation and less favourable long-term outcomes in MS. Importantly, this concordance reinforces the biological and clinical relevance of the identified clusters, suggesting that they capture distinct disease phenotypes characterised by both microbiological and clinical heterogeneity.

Discussion

In this study, the gut microbiota composition of pwMS at the onset was analysed, considering Bacteria, Archaea, and Eukarya. The aim was to identify the factors that influence its composition and to assess whether a specific subgroup of microorganisms could serve as a potential biomarker at MS onset, indicating a more favourable or unfavourable disease prognosis.

When we assessed overall microbial diversity in pwMS and HD by evaluating alpha and beta diversity metrics, we observed a significantly increased Observed richness index in pwMS compared to HD, while no significant differences were detected for Shannon and Simpson indices. These findings are consistent with previous studies reporting heterogeneous and sometimes discordant patterns in alpha and beta diversity in MS, including increased richness without changes in evenness, variable trends across disease stages, or no significant differences depending on clinical status.[45–48] However, compared to these studies, our results show the gut microbiome across the three domains at the earliest stage of MS, independent of DMT exposure and disease duration. The methodological approaches used in this study are designed to identify microbes that reflect differences associated with specific clinical features adjusting the results by the anamnestic covariates.

From our analysis, the bacterial species *Romboutsia Ilealis, Mediterraneibacter [Ruminococcus] torques,* and *C. aerofaciens* were found to be increased in pwMS compared to HD. Among these taxa, *C. aerofaciens* was more abundant in RRMS[11,49] and positively correlated with the EDSS score in patients with MS, suggesting a possible association between this bacterial species and disease severity,[50] while *Ruminococcus torques* have been associated with increased risk of MS in another study.[51] Regarding modulated Archaea, we found *Methanobrevibacter sp. TLL-48-HuF1.* Although Archaea are typically associated with extreme environments, several studies have reported their presence in the human gut, highlighting a lesser known but increasingly recognised component of the gut microbiome. Gut archaeome has been estimated to be low, individual dependent and correlated with dietary habits.[52] Methanogens seem to dominate the gut archaeal communities,[53] their role in pathology is understudied and MS evidence is mainly focused on *Methanobrevibacter*.[11,54] The human gut mycobiota displays low faecal fungal diversity together with a high degree of both within- and between-subject variability, likely reflecting the transient nature of many fungal taxa and their link with the dietary habits.[55] Looking at the differentially abundant eukarya between HD and pwMS, *K. phaffii* and *Saccharomyces cerevisiae* resulted to be increased in pwMS. However, the interpretation of these findings should be approached with caution, as the effect of fungi on human health remains controversial. Among these taxa, *S. cerevisiae* has some prior relevance to MS, as alterations in *Saccharomyces* abundance and immune correlations have already been reported in mycobiome studies, although the direction of change has not been consistent across cohorts.[12,56,57]

Functional metagenomic profiling analysis highlighted differentially abundant pathways in pwMS

percentages reported in square brackets next to the PCoA axes indicate the proportion of the total variation in the distance matrix represented by each principal coordinate. **C**. DAS resulted from comparing GC-treated and GC-untreated pwMS. Radar plot shows mean read count of DAS obtained from the comparison between GC-treated and untreated pwMS. **D**. The 46 DAS associated with clinical variables (Les = lesion burden, Gad = gadolinium-enhancing lesions, ST = subtentorial lesions, and SC = spinal cord lesions) are shown. Heatmaps display species linked to each clinical feature, identified through differential abundance analysis (top) or differential prevalence analysis (bottom) using MaAsLin3. Colours represent model coefficients for each variable, while numbers indicate statistically significant p-values derived from MaAsLin3. Empty cells (i.e., without p-values) denote non-significant association for that clinical variable. **E**. Boxplots illustrate abundance comparisons for a subset of the 46 DAS that are significantly associated with clinical variables and also consistently differ between HD and MS. P-values are derived from MaAsLin3 analysis, computed by two-sided t-test in case of abundance association or two-sided Wald z-test in case of prevalence association.

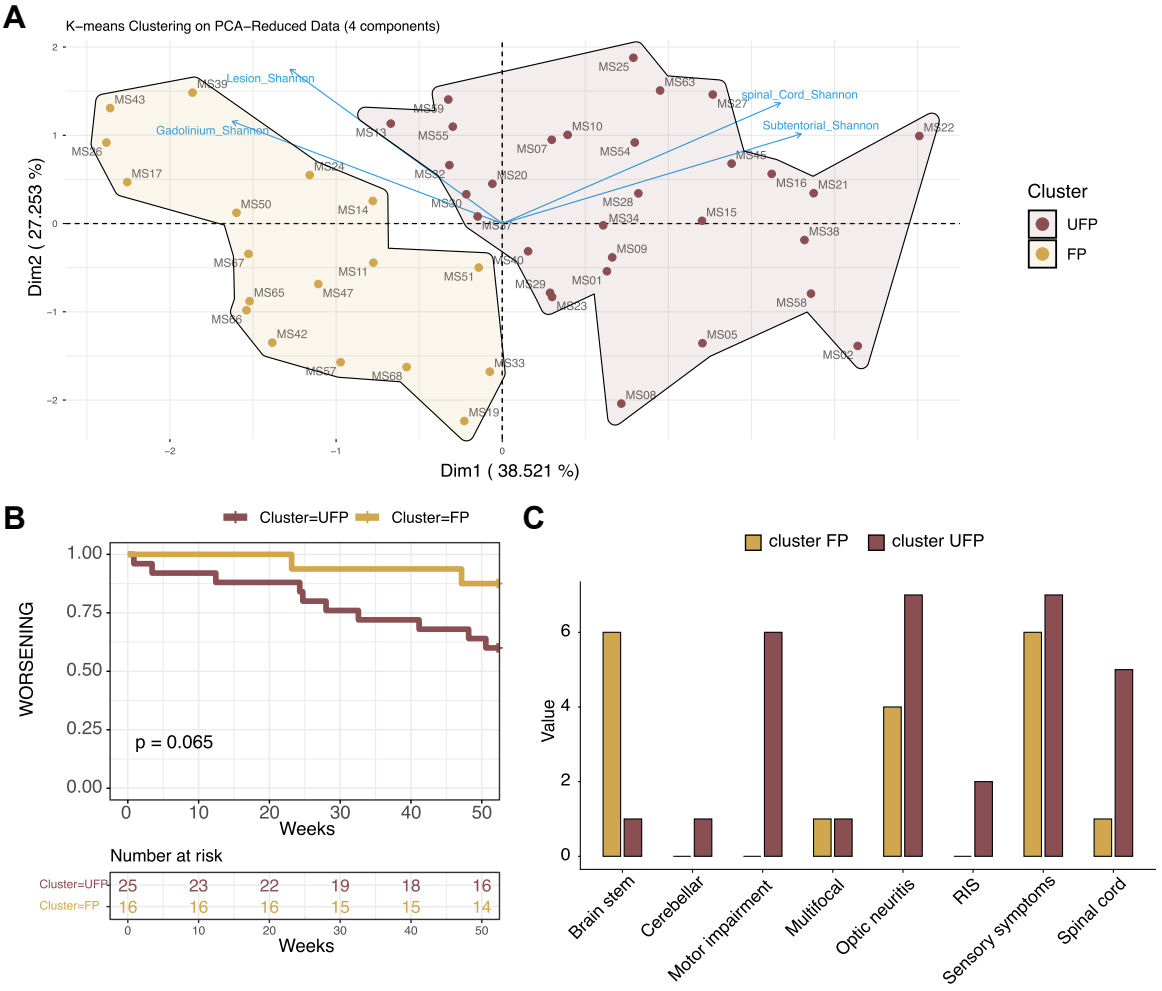


Fig. 5: **Bacterial profile can discriminate between patients with favourable (n = 16) and unfavourable (n = 25) prognosis. A**. The PCA scatter plots provide a visual representation of patient distribution within the reduced feature space, with each point corresponding to an individual pwMS projected onto the first three principal components, which together capture the majority of variance in the data. Feature loadings are shown to indicate the contribution of each variable to the principal components. **B**. Kaplan–Meier survival curves based on a two-cluster stratification of pwMS, showing a Favourable Prognosis (FP) group with a lower risk of early progression compared to the UnFavourable Prognosis (UFP) group (log-rank p = 0.065). **C**. Bar plots summarising the distribution of pwMS according to clinical presentation across FP and UFP clusters. Baseline clinical characteristics at diagnosis differed significantly between clusters (Fisher's exact test, p = 0.03).

compared to HD. Interestingly, among the microbial pathways reduced in pwMS, the decrease in succinate fermentation to butanoate pathway and L-histidine degradation I are particularly relevant, as both have immunometabolic and CNS properties and were already described in MS.[49,58]

The clinical variable most associated with variations in microbial diversity is GC treatment. GCs are a type of steroids commonly used to treat acute relapses in MS and other diseases for their immunosuppressive and anti-inflammatory properties. However, their effect on the microbiota remains only partially explored in humans. Effects of prednisone on gut microbiota have been described in epilepsy induced in rats and in mice with systemic lupus erythematosus (SLE).[59,60] A study conducted in murine models found that dexamethasone treatment induces an increase in *Bifidobacterium*, *Actinobacteria* (or *Actinomycetota*), and *Lactobacilli* in inflammatory bowel disease (IBD).[61] Conversely, in our study *P. dorei and P. clara* appear increased in GC-untreated pwMS, all belonging to the Bacteroidota phylum, while *Streptococcaceae (S. salivarius and S. vestibularis), Sutterellaceae (S. wadsworthensis, D. massiliensis) and Veillonellaceae (i.e., V. dispar, V. nakazawae, V. parvula)* were found decreased by GC-treatment. Prednisone treatment also affects the composition of the gut mycobiota, leading to a specific reduction in *Triangularia mangenotii* and *Ciliophora* species, while

increasing *Aspergillus glabripes*. Lastly, prednisone was reported to disrupt the ecological interactions between the gut mycobiome and bacteriome.[62] Despite this, our study did not identify any differential species between GC-treated and GC-untreated pwMS. Although there is still limited information in the literature, it is evident that GC can, to some extent, alter gut microbiota composition.

Our analysis, adjusted for GC treatment as a covariate, revealed differences in gut microbiota composition. However, GC-induced modulation of the gut microbiota in MS has not yet been clearly described in the literature, and longitudinal within-subject studies conducted before and after GC treatment are needed to characterise the microbial changes, their magnitude, and their duration.

Once the influence of GC treatment on microbiota diversity was established, we aimed to determine whether specific taxa were associated with different MS-related clinical traits. Specifically, the considered clinical traits included the lesion load, the presence or absence of spinal cord and infratentorial lesions, and the presence or absence of gadolinium-enhancing lesions, all recognised as negative prognostic factors for MS.[17] Clinical feature-associated DAS showed minimal overlap. This suggests that specific taxa may play a role in disease modulation across phases and clinical features of the disease. In our cohort, *B. ovatus* and *B. cellulosilyticus* were depleted in pwMS relative to HD, with further reductions of *B. ovatus* in the high-lesion-burden subgroup and of *B. cellulosilyticus* in pwMS with subtentorial lesions; conversely, *Barnesiella sp. An22* and *Romboutsia ilealis* were enriched in pwMS and in the high-lesion-burden subgroup, whereas *Clostridium innocuum* and *Ruminococcus sp. JL13D9* were enriched in pwMS and in spinal-cord-lesion-positive patients, supporting a species-level microbial signature linked not only to case–control status but also to MRI-defined disease severity and lesion topography.

B. ovatus, which was reduced in MS and in patients with higher lesion burden, is consistent with a putatively protective commensal, as it promotes indole-3-acetic-acid-driven IL-22 production and attenuates experimental colitis,[63] and it also increases intestinal SCFAs and GABA, supporting gut-brain regulatory functions.[64] By contrast, the enrichment of *Barnesiella sp. An22* in MS/high-lesion-burden cases appears disease-associated in our dataset, but the available literature suggests that *Barnesiella* effects may be highly strain-specific rather than uniformly pro-inflammatory, because the best-characterised related lineage, *Barnesiella intestinihominis*, strengthens barrier integrity, increases IL-10, lowers TNF-α and IL-6, and was found at lower abundance in subjects with greater cerebral small-vessel disease burden.[65,66] *Romboutsia ilealis* was likewise enriched in MS and in the high-lesion-burden subgroup, yet current non-MS evidence does not support a clearly inflammatory interpretation, since experimental work shows that *Romboutsia ilealis* and its metabolite 2-oxoindole-3-acetate ameliorate obesity-associated metabolic dysfunction by modulating intestinal lipid absorption.[67] In contrast, *Clostridium innocuum* fits much better with a pro-inflammatory/pathobiont interpretation, because it was independently reported as enriched in MS in a shotgun metagenomic case–control study,[68] it is a signature bacterium of Crohn's disease "creeping fat",[69] and its secreted factors can induce IL-8 and TNF-α production.[70] Similarly, the enrichment of *Ruminococcus sp. JL13D9* in MS and in spinal-cord-lesion-positive patients is compatible with an inflammatory lineage, although direct data on *JL13D9* itself remain limited; importantly, the related species *Ruminococcus gnavus* has been reported as increased in MS.[68] Finally, the reduction of *B. cellulosilyticus* in MS and in patients with subtentorial lesions is again most consistent with loss of a protective organism, because *B. cellulosilyticus* was shown to reduce IL-8, NF-κB, TNF-α, IL-1β, and IFN-γ in epithelial/immune models and modulates GABA-, serotonin-, and dopamine-related signalling in a gut-brain system,[71] while targeted enrichment of this species alleviates colitis and clinical disease in vivo.[72] Taken together, these data suggest that the depleted *Bacteroides* taxa may reflect loss of anti-inflammatory and neuromodulatory functions, whereas *Clostridium innocuum* and the *Ruminococcus* signal are more consistent with inflammatory dysbiosis.

After providing a clear and extensive description of microbiome profiles in pwMS, we focused on the community-level structure. Specifically, we aimed to interpret microbial variation at the community level rather than relying on individual species. The list of DAS identified was contextualised by computing, for each sample, an entropy value that summarises variability in microbial profiles across patients and clinical conditions. The entropy value associated with each pwMS was then used for PCA visualisation and k-means clustering. PCA revealed patterns of similarity among patients, which were further emphasised through clustering. The resulting clusters were evaluated based on clinical worsening within 12 months of diagnosis. Kaplan–Meier survival curves separated groups, indicating prognostic relevance. Notably, the resulting clusters consistently distinguished patients with a favourable from those with an unfavourable prognosis. In contrast, no single clinical trait alone or in combination was sufficient to achieve the same level of stratification. To facilitate interactive exploration of results, we developed a Shiny application named MSExplorer. This tool allows users to inspect and interact with various analyses. For example, the application enables flexible clustering, allowing users to define the number of clusters to explore different grouping configurations. Furthermore, MSExplorer

provides a comprehensive overview of the associated clinical features, Kaplan–Meier survival curves illustrating disease progression, and a chord diagram visualising the transition in EDSS scores over time. These features provide a dynamic platform for interpreting the relationship between microbial profiles and clinical outcomes.

This concept was recently inspected in a study where 41 bacterial species were identified in association with MS worsening in a patient cohort comparable to the one analysed in our study.[73] In this study, *Phoecicola vulgatus* and *Bacteroides finegoldii* were found associated with progressor severity and status. In Schwerdtfeger et al.,[74] the gut microbiome was linked to two-year changes in MS progression. Six bacteria that are present in our microbiome profile were associated with MS clinical variables. *Phocaeicola vulgatus*, *Alistipes onderdonkii*, and *Bacteroides xylanisolvens* were associated with two-year change in EDSS, whereas *Desulphovibrio fairfieldensis and Streptococcus thermophilus* were associated to cognitive performance.[74] Despite the relevance of these findings, both studies present several substantial differences compared to this paper. First, patients were not enrolled at disease onset: samples were collected after diagnosis, and clinical parameters were monitored over a five-year follow-up period or comprised a cohort of both RRMS and Progressive MS (PMS) patients.[75] In contrast, our study identified a bacterial signature from samples collected at the time of diagnosis that allows for the predictive model to be applied at the earliest stages of the disease, a critical window for initiating appropriate therapy and managing disease progression. Another difference is the use of 16S rRNA sequencing. A strength of our study is the use of shotgun metagenomic sequencing, rather than 16S rRNA sequencing. Compared to 16S rRNA sequencing, shotgun metagenomics sequencing highlights the presence of archaeal and eukaryotic microorganisms and provides higher taxonomic resolution, allowing for the discrimination of microorganisms at the species level. The challenge of analysing species from these domains is more significant than for Bacteria, mainly due to the limited availability of reference databases. As a result, many species remain unknown, their impact on human health is still unclear, and the only well-characterised species are those already known to be pathogenic. Moreover, even for better-resolved domains such as Bacteria, changes taxonomic reclassification over time, and consequently in reference databases, can alter species-level assignments.

Further investigation of the lesser-known species that make up the gut microbiota is crucial, as they may play a fundamental role in advancing our understanding of the onset and progression of diseases. Additionally, in our study, we analysed the microbiota of pwMS at the onset of the disease, when patients had not yet been treated with DMT. The absence of confounding external factors, such as therapy, whose influence on the intestinal microbiota is well known,[8,76] provides a unique opportunity for a more accurate picture of the intestinal microbiota composition resulting from the disease itself. Furthermore, since some patients had undergone treatment with GC, we further attempted to eliminate this variable by considering only the species present in the microbiota of the GC-untreated population to identify a specific microbial signature of MS. Finally, the bacterial profiles identified at the time of diagnosis, which allows for the discrimination of disease course worsening, represents a noteworthy finding with potentially significant clinical implications.

Identifying individuals at risk of early disease worsening remains one of the significant unmet needs in modern medicine. Early biomarkers can support selecting the most effective treatment strategies for each patient, promoting the shift toward personalised medicine. However, the stringent characteristics of our MS cohort, that on one hand could be considered as a strength as it enhances internal consistency and reduces confounding, also limits the generalisability of our findings to other pwMS. So, further studies in a larger cohort including i.e. pwMS with higher BMI, obese, following particular dietary regimens, or smokers are needed to validate these results. Of note, it would be necessary to extend the follow-up period as our study determined patient progression based on a one-year follow-up which is a limitation. Indeed, considering the pwMS' clinical data collected over a longer time frame would allow for a more detailed description of their disease progression over time and, consequently, a more accurate characterisation of the microbiota ecosystem. A longer follow-up period would also allow for the identification of different types of MS, such as RRMS and PMS, and would enable the investigation of whether specific microbial signatures are associated with distinct disease manifestations. Similarly, including individuals from geographically diverse areas, may make this predictive model broadly applicable. Finally, to make the results more comprehensive, performing and including analyses of the metabolites produced by the intestinal microbiota and its functionality would be necessary. This would provide information not only on the microbiota composition but also on the potential effects of different species on the entire organism.

In conclusion, our study proposes gut microbiota analysis as a potential prognostic factor for MS at diagnosis, since we revealed a peculiar gut microbiome profile in individuals with a more severe disease course. Microbiota could represent a further negative prognostic factor to be included with well-known demographic, clinical and radiological ones. It could contribute to the therapeutic decisional process aimed to identify as soon as possible those patients at higher

risk of a negative disease course. The first therapeutic intervention plays a pivotal role in shaping the entire disease trajectory. Therefore, promptly identifying high-risk patients and initiating high-efficacy treatments without delay may be crucial to preventing long-term negative outcomes.

Contributors

S.R. and M.C conceived the study; A.M., R.R., collected and processed the samples; A.M., R.R., D.T., and F.P. performed metagenomic analysis under the supervision of I.F. ,L.C and F.C.; R.L., A.L.S., C.C., E.V., F.M., M.M., S.M., P.C., recruited patients and collected metadata; S.P., S.G.C., and R.S. analysed data and created the shiny app under the supervision of M.B and F.C.; S.R., M.C., and F.C., acquired funding; A.M, R.R, D.T., F.P., F.C., and S.R wrote the manuscript. A.M., R.R., D.T., F.P., S.P., S.G.C., R.S., I.F., L.C., M.B., S.R., and F.C. accessed and verified the underlying data. Both authors confirm that all data were accurate, complete, and consistent with the analyses presented in the manuscript. All authors contributed to the critical review and revision of the manuscript and approve its final version.

Data sharing statement

The metagenomic sequencing data are available at the NCBI Sequence Read Archive with Bioproject accession number PRJNA1245671.

This study did not involve the development of novel computational tools. All methods employed are based on previously published or publicly available approaches, appropriately cited in the manuscript. The analysis workflow is available on GitHub: https://github.com/qBioTurin/MSexplorerWorkflow. The MS-Explorer app, which allows for exploring the results, is available at https://qbiogroup.shinyapps.io/MSexplorer, and on GitHub at https://github.com/qBioTurin/MSexplorer.

Declaration of interests

AM has to disclose consulting fees from Merck; RL has to disclose consulting fees, honoraria, payment for expert testimony, support for attending meetings, participation on advisory board from Bristol Meyer Squibb, Novartis, Merck, Alexion, Roche, Sanofi, Biogen, Amgen; MC received grants from Merck, Biogen, Bristol Meyer Squibb, Novartis, Roche and has to disclose consulting fees, honoraria, support for attending meetings from Merck, Sanofi, Bristol Meyer Squibb, Roche, Novartis, Biogen, Almiral; SR has to disclose consulting fees from Merck and Novartis. The remaining authors declare no conflict of interests.

Acknowledgements

We would like to thank all the volunteers in our cohort. This work was supported by Fondazione Italiana Sclerosi Multipla (FISM) grant n° 2016/R/24, by the Cassa di Risparmio di Torino Foundation, grant n° 2025.0923, and PRIN 2022 project MEDICA: Modelling and vErification of alkaptonuria and multiple sclerosis Driven by biomedICAl data [No 2022RNTYWZ]. S.R. is a professor assistant for the NODES project, which has received funding from the MUR – M4C2 1.5 of PNRR with grant agreement no. ECS00000036. We acknowledge the contribution and support from the Italian national Node (MIRRI-IT) of the European Research Infrastructure MIRRI-ERIC.

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.ebiom.2026.106470.

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